

DATA ANALYST INTERNSHIP

Task 5:

Exploratory Data Analysis (EDA)

Data Visualization & Analysis: Titanic Dataset

**Prepared by:
Balachandru S
Intern at Elevate Labs**

Data Visualization & Analysis: Titanic Dataset

1. Exploratory Data Analysis (EDA)

Objective: Perform a visual analysis of key variables in the Titanic dataset to understand relationships and draw insights regarding survival rates.

Visualizations:

A. Histogram: Age Distribution

- **Observation:**
 - The **Age** distribution is **right-skewed**.
 - The majority of passengers were **adults**, primarily between **20 to 40 years old**.
 - The distribution of children and elderly passengers is sparse.
 - **Peak in 0-10 years** shows many children were aboard.

B. Boxplot: Fare by Passenger Class

- **Observation:**
 - **Pclass 1 (First Class)** has a **highly varied fare range**, with a few **outliers** paying significantly higher fares.
 - **Pclass 3 (Third Class)** shows a **concentrated low fare** range, indicating more budget-conscious passengers.
 - The median fare for **Pclass 3** is much lower than **Pclass 1** and **Pclass 2**, reflecting socioeconomic differences.

C. Scatterplot: Age vs Fare (colored by Survival)

- **Observation:**
 - Younger passengers (below 10 years old) and **elderly passengers in First Class** have a **higher chance of survival**.
 - **Lower-class passengers (Pclass 3)** who paid less had a **lower survival rate**.
 - Higher **fare** and **First Class** are correlated with a **higher survival chance**.

Key Insights from Barplots:

A. Survival by Gender

- **Observation:**
 - **Female passengers** had a significantly **higher survival rate** compared to males.
 - The "**women and children first**" policy likely influenced this outcome.

B. Survival by Class

- **Observation:**
 - **First Class passengers** had the **highest survival rate**.
 - **Second Class** had a moderate survival rate, while **Third Class passengers** had the **lowest survival rate**.
 - **Social class** and **access to lifeboats** were crucial factors in survival, with wealthier passengers having a better chance.

Correlation Analysis:

A. Correlation Matrix Overview

The correlation matrix reveals the relationships between numeric features in the dataset.

- **Strong Positive Correlations:**
 - **SibSp and FamilySize:** As expected, these are highly correlated because $\text{FamilySize} = \text{SibSp} + \text{Parch} + 1$.
 - **Parch and FamilySize:** Similarly, family size increases with the number of parents/children aboard.
- **Moderate Correlations:**
 - **Fare and Pclass:** There is a **negative correlation** between fare and passenger class. **First-class passengers** paid higher fares, while **Third-class passengers** paid significantly less.
 - **Fare and Survived:** **Positive correlation** — higher fares are associated with higher survival rates.
- **Weak or No Correlation:**
 - **PassengerId** has no correlation with other features, as it is just an index.
 - **Age** shows a **weak correlation** with survival — age alone isn't a strong survival predictor.

Summary of Findings:

1. Survival Influences:

- **Gender:** Females had a much higher survival rate, highlighting the "women and children first" policy.
- **Passenger Class:** First-class passengers had the best chances of survival, while third-class passengers were the most vulnerable, likely due to limited access to lifeboats.
- **Fare:** Passengers who paid higher fares were more likely to survive, reflecting better living conditions and priority access to lifeboats.

2. Age and Family Size:

- **Children** (and some elderly first-class passengers) were prioritized for survival.
- Family structure (e.g., number of siblings/spouses, parents/children) has a significant correlation with family size, which may have contributed to survival decisions.

3. Class and Socioeconomic Factors:

- Wealth and class status significantly affected survival chances, with wealthier passengers in higher classes being more likely to survive.

4. Feature Relationships:

- The correlation matrix confirms that **class, fare, and family size** are strongly related, influencing the survival outcomes.

Conclusion:

The Titanic dataset reveals critical insights into the **socioeconomic disparities** that influenced survival outcomes. Key factors include **gender, class, and fare** paid, with females and wealthier passengers having a better chance of survival. Family structure also played a role in determining survival, with those in First Class being prioritized during the evacuation.